FE Graded Quiz M6

Question 1

Which one of the following models is **not** a version of the Dickey Fuller test?

- An AR(1) model with drift and time trend
- An AR(1) model without drift nor time trend
- An AR(1) model with drift but no time trend
- An AR(1) model with an interaction term of drift and time trend

Question 2

Assume we have the following vector error correction model:

$$\Delta X_t = C + \Pi X_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta X_{t-i} + \epsilon_t$$

Where X_t is a time series vector, ϵ_t is a white noise vector, and C is a deterministic vector.

Which term is used to model short-term dynamics?

- $\sum_{i=1}^{p-1} \Gamma_i \Delta X_{t-i}$
- ΠX_{t-1}
- $\Pi X_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta X_{t-i}$
- $C + \Pi X_{t-1}$

Question 3

Which of the following conditions can ensure x_t is ergodic for the mean?

TOP

- x_t is normally distributed and $\sum_{j=0}^{\infty} cov(x_t, x_{t-j}) < 0$
- x_t is non-stationary and $\sum_{j=0}^{\infty} cov(x_t, x_{t-j}) < \infty$
- x_t is normally distributed and $\sum_{j=0}^{\infty} cov(x_t, x_{t-j}) < \infty$
- x_t is stationary and $\sum_{j=0}^{\infty} cov(x_t, x_{t-j}) < \infty$

Question 4

If two time series are non-stationary and cointegrated, which model can we choose to model their interaction?

- VAR model
- Error correction model
- Ridge regression
- OLS regression

Question 5

According to the following Augmented Dickey Fuller test result, which of the following statements is correct?

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression trend


Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)

Residuals:
       Min        1Q    Median        3Q       Max
-0.0308928 -0.0031066 -0.0000308  0.0030585  0.0231592

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.938e-03  3.451e-03   2.300   0.0216 *
z.lag.1     -6.966e-03  3.089e-03  -2.255   0.0243 *
tt           8.924e-08  3.249e-07   0.275   0.7836
z.diff.lag  -1.411e-02  2.583e-02  -0.546   0.5850
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.005059 on 1499 degrees of freedom
Multiple R-squared:  0.003942,  Adjusted R-squared:  0.001949
F-statistic: 1.978 on 3 and 1499 DF,  p-value: 0.1154


Value of test-statistic is: -2.2554 1.8507 2.7381

Critical values for test statistics:
      1pct  5pct 10pct
tau3 -3.96 -3.41 -3.12
phi2  6.09  4.68  4.03
phi3  8.27  6.25  5.34
```

- There is no unit root.
- There is a unit root and trend should be excluded.
- There is a unit root, and drift and trend should be excluded.

TOP

- There is a unit root, and drift should be excluded.

Question 6

Assume we have the following vector error correction model:

$$\Delta X_t = C + \Pi X_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta X_{t-i} + \epsilon_t$$

Where X_t is a time series vector, ϵ_t is a white noise vector, and C is a deterministic vector.

If Π has full rank K, which of the following statements is **correct**?

- All time series in X_t have unit roots.
- All time series in X_t vector can be used in a VEC model.
- All time series in X_t vector are cointegrated.
- All time series in X_t vector are stationary.

Question 7

What is an estimation method that we use to estimate a VAR model?

- OLS estimation
- Bayesian estimation
- Non-linear estimation
- Non-parametric estimation

Question 8

If the U.S. 30-year interest rate and U.S. 10-year interest rate are cointegrated, which of the following statements is correct?

TOP

- The U.S. 30-year interest rate and U.S. 10-year interest rate are independent.
- U.S. 30-year interest rate and U.S. 10-year interest rate have a long-term stable relationship.
- The difference of the two rates is non-stationary.
- The U.S. 30-year interest rate and U.S. 10-year interest rate are both stationary time series.

Question 9

Which of the following statements is **correct** for $x_t \sim I(3)$?

- x_t can be called integrated with order two.
- $(1-B)^3 y_t$ is not a stationary time series.
- Augmented Dickey Fuller test on x_t will reject H_0.
- $(1-B)^3 y_t$ is a stationary time series.

Question 10

Which of the following can be a maximum likelihood test for cointegration when r is the rank of the coefficient matrix of lagged time series in a VEC model?

- H_0: r> 0/H_1: r=1
- H_0: r= 0/H_1: r> 1
- H_0: r= 0/H_1: r= 1
- H_0: r> 0/H_1: r> 1

Question 11

When some of the time series in a VAR model have trend components,

TOP

which of the following methods is **not** the right way to construct a VAR model?

- Use the original form of time series to build a VAR model.
- Run a regression of time series on trend and use residuals to build a VAR model.
- De-trend the time series with a trend component and then build a VAR model.
- Add trend variable to a VAR model.

Question 12

Which of the following is **not** a common feature for spurious regression?

- Low R^2
- The absolute value of coefficient from the regression is high
- Low DW test value
- t-test is significant

Question 13

Which of the following results shows that time series x_t is stationary?

- ADF test cannot reject H_0 and KPSS test cannot reject H_0
- ADF test rejects H_0 and KPSS test rejects H_0
- ADF test cannot reject H_0 and KPSS test rejects H_0
- ADF test rejects H_0 and KPSS test cannot reject H_0

Time left: 59:14

QUESTIONS

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

WORLDQUANT UNIVERSITY

TOP

Question 14

Assume we have the following vector error correction model:

$$\Delta X_t = C + \Pi X_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta X_{t-i} + \epsilon_t$$

Where X_t is a time series vector, ϵ_t is a white noise vector, and C is a deterministic vector.

Π can be split into which of the following two components?

- Cointegrating vectors and penalty vectors
- Stationary vectors and adjustment vectors
- Cointegrating vectors and adjustment vectors
- Stationary vectors and penalty vectors

Question 15

A VAR model is **not** a special case of which of the following model groups?

- GARCH
- VARMA
- ARMAX
- AR

Question 16

If x_t and y_t are cointegrated, what property does the coefficient estimate have from the regression of x_t on y_t?

- Quasi-consistent
- Super-consistent
- Biased

TOP

Inefficient

SUBMIT

WORLDQUANT UNIVERSITY

TOP